Acid/Metal Reactions

Hydrochloric acid mixed with magnesium.

$2\ HCl + Mg \rightarrow MgCl_2 + H_2$

Hydrochloric acid mixed with lithium metal.

$2\ HCl + 2\ Li \rightarrow 2\ LiCl + H_2$

Hydrochloric acid mixed with calcium metal.

$2\ HCl + Ca \rightarrow CaCl_2 + H_2$

Nitric acid mixed with potassium.

$2\ HNO_3 + 2\ K \rightarrow 2\ KNO_3 + H_2$

Neutralisation

1. Write the reactants
2. Write the products (start with the easy ones like H_2 or H_2O).
3. Work out the formula for the salt.
4. Balance the equation.

Hydrochloric acid mixed with potassium hydroxide.

$HCl + KOH \rightarrow KCl + H_2O$

Nitric acid mixed with sodium hydroxide.

$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$

Hydrochloric acid mixed with magnesium hydroxide.

$2\ HCl + Mg(OH)_2 \rightarrow MgCl_2 + 2\ H_2O$

Nitric acid mixed with calcium hydroxide.

$2\ HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2\ H_2O$

Sulfuric acid mixed with calcium hydroxide.

$H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2\ H_2O$

Acid/Carbonate Reactions

Hydrochloric acid mixed with calcium carbonate.

$2\ HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$

Hydrochloric acid mixed with barium carbonate.

$2\ HCl + BaCO_3 \rightarrow BaCl_2 + H_2O + CO_2$